



HOT WORK SAFETY

Any work that involves welding, cutting, grinding, or any activity that produces heat, sparks, or open flames.

HAZARDS

-  Fire and explosion
-  Flammable gases, vapors, and liquids
-  Sparks igniting combustible materials
-  Burns and heat stress
-  Toxic fumes and smoke inhalation
-  Eye injuries from UV light, sparks, or hot metal
-  Oxygen-enriched environment
-  (increased fire risk)
-  Poor ventilation
-  Noise from cutting and grinding
-  Lack of fire watch or inadequate supervision

PRECAUTIONS

-  Obtain and follow a valid Hot Work Permit
-  Inspect work area and remove combustible materials
-  Use fire-resistant blankets/barriers
-  Provide adequate ventilation and gas monitoring 
-  Use proper PPE:
 - Face shield, safety goggles
 - Heat-resistant gloves
 - Flame-resistant (FR) clothing
 - FFP3 respirators for fumes and smoke
-  Keep fire extinguishers readily available
-  Assign a trained Fire Watch during and after work
-  Limit noise exposure and use hearing protection



LIFTING OPERATIONS SAFETY

Any operation using cranes, hoists, or lifting equipment to move loads.

HAZARDS

-  Crane overturn or collapse
-  Falling loads / dropped objects
-  Swinging or unstable loads
-  Equipment failure (wire rope, hook, sling)
-  Pinch injuries from UV light, sparks, or hot metal
-  Oxygen-enriched environment
-  Unauthorized personnel in the area
-  Weather conditions (wind, poor visibility)
-  Inadequate training or certification

PRECAUTIONS

-  Obtain and follow a valid Lifting Permit / Plan
-  Use certified lifting equipment & accessories (SWL clearly marked) 
-  Conduct pre-use inspection of crane, slings, hooks, and shackles
-  Keep unauthorized persons out of lifting zone
-  Assign a trained Banksman / Signaller
-  Maintain clear communication (hand signals, radios)
-  Monitor weather conditions and stop work in high winds
-  Provide load properly with slings, shackles, and tag lines
-  Establish and communicate a clear Emergency/ Rescue Plan



TOOLBOX TALK SAFETY

Short safety meetings conducted before starting work to raise awareness about hazards and precautions

HAZARDS

- ! Workers unaware of job-specific hazards
- ! Miscommunication between team members
- ! Ignoring Permit-to-Work requirements
- ! Incomplete understanding of lifting, hot work, or confined space risks
- ! Lack of awareness about emergency procedures
- ! Use of equipment without proper briefing
- ! Workers not wearing correct PPE
- ! Failure to identify changes in worksite conditions
- ! Complacency due to routine work

PRECAUTIONS

- ✓ Conduct daily toolbox talks before starting tasks
- ✓ Focus on specific hazards related to the job of the day
- ✓ Ensure Permit-to-Work requirements are explained
- ✓ Encourage worker participation and feedback
- ✓ Use visual aids or demonstrations for clarity
- ✓ Confirm all workers understand emergency procedures
- ✓ Reinforce correct use of PPE and equipment
- ✓ Highlight lessons learned from recent incidents/near misses
- ✓ Keep talks short, clear, and interactive



WORKING AT HEIGHT

Any work where a person could fall from one level to another and cause injury (e.g., ladders, scaffolds, rooftops).

HAZARDS

-  Falls from ladders, scaffolds, rooftops
-  Falling objects / tools
-  Weak or unstable surfaces (wind, rain, heat)
-  Lack of edge protection / guardrails
-  Slips, trips, and missteps
-  Incorrect use of PPE (harness, lifeline)
-  Electrical hazards (overhead lines)
-  Fatigue or lack of concentration
-  Ensure workers hold valid Working at Height certifications
-  Ensure a dedicated Rescue Team on standby
-  Prepare and commun-

PRECAUTIONS

-  Use fall protection system (harness, lifeline, anchor points)
-  Inspect ladders, scaffolds, and PPE before use
-  Ensure guardrails and toe boards are in place
-  Keep working surfaces clean and dry
-  Secure tools and materials to prevent falling objects
-  Avoid working at height in bad weather
-  Provide training & supervision for workers
-  Ensure proper lighting and visibility
-  Maintain clear communication system
-  Follow Permit-to-Work system for high-risk tasks
-  Ensure workers hold valid Working at Height certifications



ELECTRICAL SAFETY

Any work involving electrical systems, tools, or equipment.

HAZARDS

-  Electric shock
-  Electrocution
-  Burns from arc flash / arc blast
-  Fire from faulty wiring or overloaded circuits
-  Explosion due to flammable atmosphere
-  Contact with overhead or underground power lines
-  Faulty or damaged cables and plugs
-  Wet conditions increasing shock risk
-  Inadequate grounding
-  Unauthorized or unqualified personnel handling electricity

PRECAUTIONS

-  Obtain and follow a valid Permit to Work for electrical activities
-  Only qualified & certified electricians to perform electrical tasks
-  Inspect cables, plugs, and equipment before use
-  Ensure proper grounding and insulation
-  Use lockout/tagout (LOTO) procedures before maintenance
-  Maintain safe clearance from overhead power lines
-  Avoid using electrical equipment in wet or damp
-  Use correct PPE (insulated gloves, safety boots, arc-flash clothing)
-  Provide Residual Current Devices (RCDs) where required



HAND TOOLS & POWER TOOLS SAFETY

Safe use of manual and powered tools to prevent injuries and accidents at worksites

HAZARDS

- ⚠ Cuts, punctures, and abrasions from sharp tools
- ⚡ Flying particles during cutting, grinding, or drilling
- ⚡ Electric shock from faulty power tools
- ⚠ Noise and vibration leading to hearing loss or hand-arm vibration syndrome
- 🔥 Burns or sparks from grinding and welding tools
- ⚠ Fire due to overheating or faulty wiring
- ⚠ Eye injuries from lack of goggles/face shield
- 🚶 Slips, trips, and falls caused by misplaced tools or cables
- ⚠ Struck-by hazards from dropped or loose tools
- ⚠ Using damaged, non-certified, or makeshift tools
- ⚠ Fatigue from improper ergonomic use

PRECAUTIONS

- ✓ Inspect all tools before use; remove damaged ones from service
- ✓ Use correct PPE: safety goggles, gloves, hearing protection, face shield 
- ✓ Keep cutting edges sharp and tools in good condition
- ✓ Ensure power tools are properly grounded insulated
- ✓ Disconnect tools when not in use or during maintenance
- ✓ Secure loose clothing, hair, and jewelry before operating tools
- ✓ Maintain clean, organized work areas to avoid trips and falls
- ✓ Use guards in safe handling and storage
- ✓ Limit exposure to noise and vibration with breaks and protection



HOUSEKEEPING SAFETY

Maintaining a clean and organized workplace to reduce risks of accidents and injuries.

HAZARDS

-  Slips, trips, and falls due to clutter or sls
-  Blocked emergency exits or escape routes
-  Accumulation of combustible waste (fire hazard)
-  Poor storage ok tion leading to respiratory hazards
-  Sharp objects or debris causing cuts & puncture
-  Poor visibility due to unorganized worksite
-  Increased risk of pest infestation from waste
-  Struck-by hazards from misplaced equipment
-  Struck-by hazards from misplaced equipment
-  Fatigue and stress due to messy workilace

PRECAUTIONS

-  Keep all work areas clean and free of clutter
-  Immediately clean up spills of oil, water, or chemicals
-  Ensure all emergency exits and escape routes are clear
-  Store tools and materials in designated areas
-  Remove combustible waste regularly and safely
-  Use proper containers for sharp or hazardous waste
-  Provide adequate lighting in all work areas
-  Conduct regular housekeeping inspections
-  Train workers on housekeeping best practices
-  Establish and communicate a Housekeeping Policy



LINE OF FIRE SAFETY

Situations where workers are positioned in path of moving objects, equipment, or energy release that can cause injury.

HAZARDS

- ⚠ Being struck by moving vehicles or equipment
- ⚠ Caught-between hazards (workers between equipment, walls, or materials)
- ⚠ Falling objects from cranes, scaffolds, or shelves
- ⚠ Release of stored energy (pressurized systems, hydraulic lines)
- ⚠ Flying particles from grinding, cutting, or drilling
- ⚠ Unexpected equipment startup during maintenance
- ⚠ Collapse of unstable materials or structures
- ⚠ Poor awareness of exclusion zones
- ⚠ Slips, trips and falls into hazardous areas
- ⚠ Lack of barriers or warning signs

PRECAUTIONS

- ✓ Identify and mark all line of fire hazards before work
- ✓ Establish and enforce exclusion zones ++ around high-risk areas
- ✓ Keep workers clear of suspended or moving loads
- ✓ Provide spotters / banksmen during equipment operations
- ✓ Use proper PPE (helmets, safety boots, goggles, hi-vis vests)
- ✓ Secure tools and materials to prevent falling objects
- ✓ Release stored energy safely before maintenance (LOTO)
- ✓ Use guards and barriers where possible
- ✓ Ensure clear communication between workers and opera-
- ✓ Establish and communicate an Emergency Response Plan



DRIVING SAFETY

Safe driving practices for company vehicles, heavy equipment, and personal cars inside works

HAZARDS

PRECAUTIONS



Speeding or reckless driving inside worksite



Driver fatigue or distraction (mobile phone use, eating, stres)



Collisions with workins, equipment, or other vehicles



Poor weather conditions (fog, dust, rain)



Blind spots durring reversing or maneuvering



Vehicle overturn due uneven or soft ground



Lack of seat belt use



Unauthorized or untrained drivers operating vehicles



Brake, tire, or light failure duto poor maintenance



Driving under ine influenc of alcohol, drugs, or medication



Enforce strict speed limits



Prohibit mobile phone use and distractions while driving



Ensure all drivers are trained, licensed, and authorized



Mandatory seat belt use for all occupants



Conduct daily vohicle inspections (brakes, tires, lights, mirrors)



Restrict driving in unsafe weather or poor visibility conditions



Provide adequate lighting and signages on site roads



Use reversing alarms, pfevent griverfatigue



Establish and communate an Emergency / Accident Response Plan



EXCAVATION & TRENCHING SAFETY

Work involving digging, trenching, or excavation near pipelines, cables, or underground utilities.

HAZARDS

-  Collapse of trench walls (cave-ins)
-  Contact with underground utilities (gas, electricity, water)
-  Falling objects or materials into excavation
-  Hazardous gases or oxygen deficiency in deep excavations
-  Flooding from groundwater or rain
-  Workers or equipment falling into trench
-  Heavy equipment operating too close to edge
-  Vibration from traffic causing collapse
-  Poor access/egress (no safe ladders or ramps)
-  Unadequate shoring, shielding, or trench support
-  Lack of inspection by competent person

PRECAUTIONS

-  Obtain and follow a valid Excavation Permit.
-  Identify and mark all underground utilities before digging
-  Provide proper shoring, shielding, or benching
-  Keep excavated materials at least 1 meter from trench edge
-  Test atmosphere for hazardous gases or oxygen levels
-  Provide safe access/egress every 7.5 m
-  Install barriers, fencing, and warning signs
-  Control water accumulation (pumps, drainage systems)
-  Maintain safe distance of heavy equipment from edges
-  Conduct frequent inspections by competent person
-  Provide proper PPE and an Emergency/Rescue Plan



FIRE SAFETY

Any activity or workplace condition that may cause or contribute to fire risks.

HAZARDS

PRECAUTIONS



Presence of flammable liquids, gases, or solids



Identify and store flammable materials safely



Electrical faults causing sparks or overheating



Install and maintain fire detection and alarm systems



Hot work without permit or controls



Provide adequate and accessible fire extinguishers (CO₂, foam, powder)



Accumulation of combustible waste/materials



Conduct regular fire drills and emergency training



Overloaded circuits or faulty wiring



Apply Hot Work Permit system for welding, grinding, or cutting



Lack of fire detection and alarm systems



Avoid overloading electrical systems and ensure maintenance



Blocked or inadequate fire exits



Maintain good housekeeping to reduce fire hazards



Absence of firefighting equipment (extinguishers, etc.)



Provide proper PPE (FR clothing, gloves, goggles) for hot work



Poor housekeeping increasing fire load



Smoking in restricted areas



Lack of fire drills and emergency preparedness



COMPRESSED GAS CYLINDERS SAFETY

Safe handling, storage, and use of
compressed gas cylinders

HAZARDS

PRECAUTIONS

- | | |
|---|--|
|  Cylinder explosion due to dropping or damage |  Store cylinders upright secured with chains or brackets |
|  Toxic or flammable gas leaks |  Keep away from heat, ignition sources, and electrical circuits |
|  Fire or explosion near ignition sources |  Use correct regulators, hoses, and nozzles |
|  Cylinder "rocket" if valve breaks |  Check for leaks before use |
|  Frostbite from gas discharging |  Close valves after work completion |
|  Mixing of incompatible gases (e.g. O ₂ and fuel gas) |  Separate full and empty cylinders with clear labeling |
|  Regulator or hose malfunctions |  Isolate oxygen from flammable gas |
|  Poor storage or inadequate ventilation |  Wear PPE (gloves, goggles, safety boots) |
|  Unlabeled or unknown cylinders |  Use trolleys, not rolling to move cylinders |
|  Unsafe handling or by untrained persons |  Provide adequate ventilation in storage and work areas |

PERMIT TO WORK SAFETY

A formal system used to control high-risk activities and ensure all hazards are identified and controlled before work begins.

HAZARDS

- ⚠ Unauthorized work without permits
- ⚠ Workers unaware of site-specific hazards
- ⚠ Incorrect isolation of energy sources
- ⚠ Incomplete risk assessment before starting tasks
- ⚠ Simultaneous operations creating conflicts (SIMOPS)
- ⚠ Hot work, confined space, lifting, or electrical tasks without control
- ⚠ Communication gaps between departments (operations, maintenance, contractors)
- ⚠ Failure to return the worksite to a safe condition after completion
- ⚠ Lack of supervision during permitted activities
- ⚠ Forged, incomplete, or expired permits

PRECAUTIONS

- ✓ Issue permits only through authorized personnel
- ✓ Conduct a thorough risk assessment / HITRA before
- ✓ Ensure all required isolations and lockouts are in place
- ✓ Communicate hazards, precautions, and responsibilities to all workers
- ✓ Verify workers hold valid certifications for specialized tasks
- ✓ Ensure SIMOPS coordination when multiple jobs overlap
- ✓ Keep a standby person / fire watch for high-risk tasks
- ✓ Regularly inspect and audit active permits on site
- ✓ Train workers and supervisors on the Permit to Work system
- ✓ Prohibit work if permit is missing, expired or incomplete
- ✓ Ensure proper closure and hand-back of continuous



HEARING CONSERVATION SAFETY

Preventing noise-induced hearing loss
among workers exposed to high noise levels

HAZARDS

- ⚠ Hearing loss due to prolonged exposure to high noise levels
- ⚠ Tinnitus (ringing in the ears)
- ⚠ Fatigue and reduced concentration caused by noise
- ⚠ Miscommunication and accidents due to poor hearing in noisy areas
- ⚠ Ear infections from improper use of hearing protection
- ⚠ Psychological stress due to continuous loud environments
- ⚠ Permanent hearing damage from explosive or sudden loud noise
- ⚠ Non-compliance with noise level regulations

PRECAUTIONS

- ✓ Conduct regular noise level monitoring in the workplace
- ✓ Provide appropriate hearing protection devices (earplugs, earmuffs)
- ✓ Ensure workers are trained on correct use of hearing PPE
- ✓ Limit time spent in high-noise areas with job rotation
- ✓ Post warning signs in designated high-noise zones
- ✓ Maintain equipment to reduce unnecessary noise
- ✓ Encourage regular hearing tests for exposed workers
- ✓ Enforce a clear Hearing Conservation Program



CHEMICAL HANDLING SAFETY

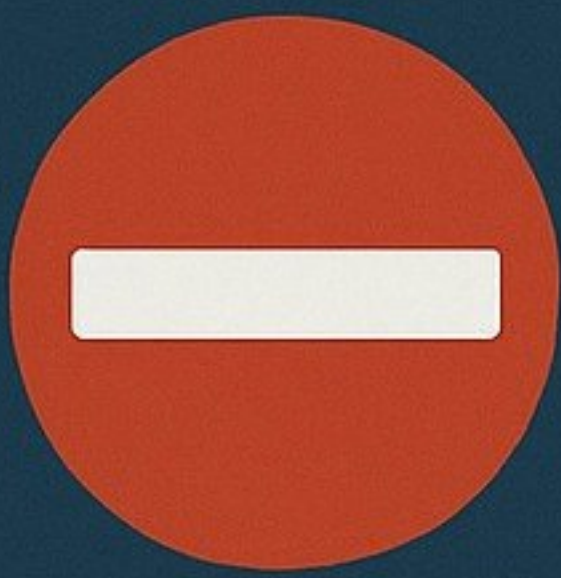
Safe storage, transfer, and use of hazardous chemicals to prevent exposure, fire, and contamination

HAZARDS

-  Inhalation of toxic vapors or gases
-  Skin contact causing burns or irritation
-  Chemical splashes to eyes or face
-  Fire and explosion from flammable chemicals
-  Corrosion of containers or pipelines
-  Mixing incompatible substances
-  Spills and leaks contaminating the environment
-  Inadequate labeling or misidentification of containers
-  Improper PPE usage or lack of training
-  Poor ventilation in work areas

PRECAUTIONS

-  Read and follow Material Safety Data Sheets (MSDS)
-  Store chemicals in labeled, approved containers
-  Keep flammable and reactive chemicals
-  Use proper PPE (gloves, goggles, aprons, FFP3 respirator)
-  Provide secondary containment (spill trays, bunds)
-  Ensure proper ventilation in storage and work areas
-  Never mix chemicals unless authorized and supervised
-  Train workers on chemical hazards and safe handling
-  Keep emergency eyewash and shower stations nearby
-  Inspect containers and storage areas regularly



BYPASSING SAFETY CONTROLS

Any intentional disabling, overriding, or ignoring of safety systems, alarms, or procedures

HAZARDS

- ⚠️ Serious injuries or fatalities due uncontrolled energy release
- ⚠️ Fire, explosion, or toxic gas release
- ⚠️ Electrocution from bypassed interlocks
- ⚠️ Equipment damage or catastrophic failure
- ⚠️ Exposure to hazardous chemicals or pressure systems
- ⚠️ Falls from height due disabled fall protection
- ⚠️ Confined space entry without gas testing/ventilation
- ⚠️ Loss of critical alarms or shutdown systems
- ⚠️ Legal, regulatory, and disciplinary consequences
- ⚠️ Negative safety culture across the workplace
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PRECAUTIONS

- ✅ Enforce a strict Zero Tolerance Policy for bypassing controls
- ✅ Ensure all safety systems and interlocks remain active
- ✅ Provide regular training and awareness about risks of bypassing
- ✅ Monitor and audit safety-critical equipment regularly
- ✅ Require management authorization before any override
- ✅ Implement Lockout/Tagout (LOTO) for maintenance tasks
- ✅ Encourage and empower Stop Work Authority
- ✅ Investigate all bypass attempts immediately and transparently
- ✅ Apply disciplinary action for violations, but reward compliance



HEAVY EQUIPMENT CROSSING OVER LIVE BURIED PIPELINES

HAZARDS

- ⚠ Damage or rupture to live buried pipelines (oil, gas, water)
- ⚠ Ground collapse due to weak or unstable soil
- ⚠ Equipment overturn or getting stuck in soft ground
- ⚠ Leakage or explosion from damaged pipeline
- ⚠ Fire or toxic gas release from ruptured line
- ⚠ Absence of valid Line Detection or UXO Clearance Certificate
- ⚠ Lack of Permit to Work (PTW) or crossing approval
- ⚠ Untrained operators unaware of underground
- ⚠ Poor communication between teams (lifting, excavation, operation)
- ⚠ Excessive ground vibration reducing buried line integrity
- ⚠ Ensure minimum safe clearance between pipeline and surface load

PRECAUTIONS

- ✓ Obtain valid Line Detection Report and UXO Clearance Certificate
- ✓ Conduct HITRA and issue PTW before crossing
- ✓ Clearly mark pipeline routes on the ground
- ✓ Install steel plates or heavy-duty mats to distribute equipment weight
- ✓ Ensure minimum safe clearance between pipeline and surface load
- ✓ Do not cross over live pipelines unless client approval is granted
- ✓ Assign spotter / banksman to guide equipment movement
- ✓ Inspect soil condition continuously during crossing
- ✓ Keep firefighting and emergency response team on standby
- ✓ Stop work immediately



PIPELINE



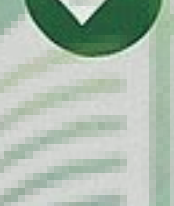
ELECTRICAL SAFETY

Any work involving electrical systems, equipment, or tools that can expose workers to electrical hazards.

HAZARDS

-  Electric shock from live wires or faulty equipment
-  Electrocution leading to fatalities
-  Burns from arc flash or arc blast
-  Fire caused by overloaded circuits or faulty wiring
-  Explosion in presence of flammable gases/vapors
-  Contact with overhead or underground power lines
-  Faulty or damaged cables, plugs, or insulation
-  Wet or damp conditions increasing shock risk
-  Inadequate grounding of equipment
-  Unauthorized or unqualified personnel performing electrical work
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PRECAUTIONS

-  Obtain and follow a valid Permit to Work for electrical tasks
-  Only qualified and certified electricians to handle electrical jobs
-  Inspect all tools, cables, and plugs before use
-  Ensure proper grounding and insulation
-  Apply Lockout/Tagout (LOTO) before maintenance or repair
-  Maintain safe clearance from overhead power lines
-  Do not use electrical tools in wet or damp conditions
-  Use correct PPE: insulated gloves, dielectric boots, arc-flash suit, face shield
-  Install and maintain Residual Current Devices (RCDs)
-  Keep CO₂ fire extinguishers available for electrical fires



TRAFFIC MANAGEMENT SAFETY

Safety measures to control vehicle and equipment movement in worksites.

HAZARDS

- ! Vehicle collisions with workers or other equipment
- ! Unauthorized vehicles entering the worksite
- ! Poor visibility (dust, night work, blind spots)
- ! Speeding or reckless driving inside the site
- ! Lack of separation between pedestrians and vehicles
- ! Reversing accidents due blind spots
- ! Equipment overturn from unstable ground
- ! Distracted driving (phone use, fatigue)
- ! Inadequate or missing traffic signs and barriers
- ! Lack of trained flagmen / banksmen

PRECAUTIONS

- ✓ Develop and implement a Traffic Management Plan
- ✓ Install clear traffic signs, cones, and barriers
- ✓ Enforce strict speed limits inside the site
- ✓ Separate pedestrian walkways from vehicle routes
- ✓ Assign trained flagmen / banksmen for guidance
- ✓ Ensure vehicles have reverse alarms and cameras
- ✓ Conduct daily vehicle inspections (brakes, tires, lights)
- ✓ Provide adequate lighting for night operations
- ✓ Monitor driver behavior (no phone use, no fatigue driving)
- ✓ Provide training and induction for all drivers/operators



CONFINED SPACE SAFETY

Any area with limited entry/exit not designed for continuous occupancy (e.g., tanks, silos, sewers).

HAZARDS

- ⚠ Oxygen deficiency
- 🔥 Toxic or flammable gases
- ⚡ Fire and explosion
- ⚠ Heat stress
- ⚠ Limited movement or entrapment
- 🔊 Noise and vibration
- ⚠ Poor visibility and inadequate lighting
- ⚠ Falling objects inside the confined space
- ⚠ Slips, trips, and falls
- ⚠ Panic or disorientation due to poor communication
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- ⚠ Panic or disorientation due to poor communication

PRECAUTIONS

- ✅ Test atmosphere before entry (O₂ gases, vapors)
- ✅ Provide ventilation and continuous gas monitoring
- ✅ Follow Permit-to-Work system for confined spaces
- ✅ Use appropriate PPE (respirators, harness, communication devices)
- ✅ Ensure a standby person with a rescue plan is present
- ✅ Isolate hazardous energy sources (LOTO)
- ✅ Provide continuous training and supervision
- ✅ Monitor temperature and humidity
- ✅ Ensure proper lighting for clear visibility
- ✅ Limit noise exposure with hearing protection
- ✅ Establish and communicate



LOTO SAFETY

HAZARDS

-  Unexpected startup during maintenance
-  Release of stored energy (electrical, hydraulic, pneumatic, thermal)
-  Electrocution from live circuits not isolated
-  Crushing, cutting, or entanglement by moving parts
-  Burns from hot surfaces or steam
-  Chemical exposure from pressurized systems
-  Failure to isolate all energy sources
-  Unlabeled or missing LOTO devices
-  Unauthorized removal of locks/tags
-  Lack of training or supervision
- 

PRECAUTIONS

-  Develop and follow a written LOTO program
-  Identify and isolate all energy sources before work
-  Use approved lockout and tagout devices for each source
-  Verify zero energy state before starting maintenance
-  Train all workers on LOTO procedures and responsibilities
-  Ensure only authorized perform lockout/tagout
-  Keep clear labeling of all switches, valves, and circuits
-  Never remove another worker's lock or tag without authorization
-  Conduct regular audits of LOTO compliance



ISOLATION OPERATIONS SAFETY

Procedures to isolate hazardous energy sources (electrical, mechanical, hydraulic, pneumatic, chemical, thermal) before maintenance or repair work

HAZARDS

- ⚠ Accidental release of stored energy
- ⚠ Unexpected startup of machinery or equipment
- ⚠ Electric shock from live circuits
- ⚠ Hydraulic or pneumatic pressure release
- ⚠ Chemical leaks or toxic exposure
- ⚠ Contact with hot surfaces or steam
- ⚠ Inadequate identification of isolation points
- ⚠ Failure of isolation devices (valves, switches)
- ⚠ Unauthorized removal of isolations
- ⚠ Lack of supervision and verification
- ⚠

PRECAUTIONS

- ✓ Develop and implement a clear isolation Procedure
- ✓ Identify and label all isolation points (switches, valves, breakers)
- ✓ Use approved isolation devices (locks, blanks, blinds, valve covers)
- ✓ Verify zero-energy state before starting work
- ✓ Isolate and lockout all hazardous energy sources (mechanical, electrical, hydraulic, pneumatic, chemical, thermal)
- ✓ Maintain a record/log of all isolation activities
- ✓ Only authorized personnel perform isolations
- ✓ Provide continuous supervision during critical tasks
- ✓ Use proper PPE (gloves, goggles, insulated tools)



EMERGENCY EVACUATION SAFETY

HAZARDS

- ⚠ Panic and stampede during evacuation
- ⚠ Blocked or locked emergency exits
- ⚠ Poor visibility due to smoke, dust, or power failure
- ⚠ Slips, trips, and falls in escape routes
- ⚠ Miscommunication with lack of clear instructions
- ⚠ Workers unfamiliar with evacuation procedures
- ⚠ Injuries while using emergency stairs or ladders
- ⚠ Failure to account for all personnel after evacuation
- ⚠ Delay in medical or rescue response
- ⚠ Inadequate drills leading to confusion in real emergencies
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- ⚠ Inadequate drills leading to confusion in real

PRECAUTIONS

- ✓ Develop and communicate a clear Emergency Evacuation Plan
- ✓ Ensure all emergency exits are marked, illuminated, and unobstructed
- ✓ Conduct regular fire and evacuation drills for all staff
- ✓ Install emergency lighting along escape routes
- ✓ Provide clear signage showing exit paths and assembly points
- ✓ Train workers on roles and responsibilities during evacuation
- ✓ Designate and train fire wardens / marshals
- ✓ Maintain updated assembly points in safe zones
- ✓ Keep a headcount / roll call system to account for
- ✓ Coordinate evacuation plans with local emergency services
- ✓ Review and update evacuation procedures



INCIDENT REPORTING & INVESTIGATION SAFETY

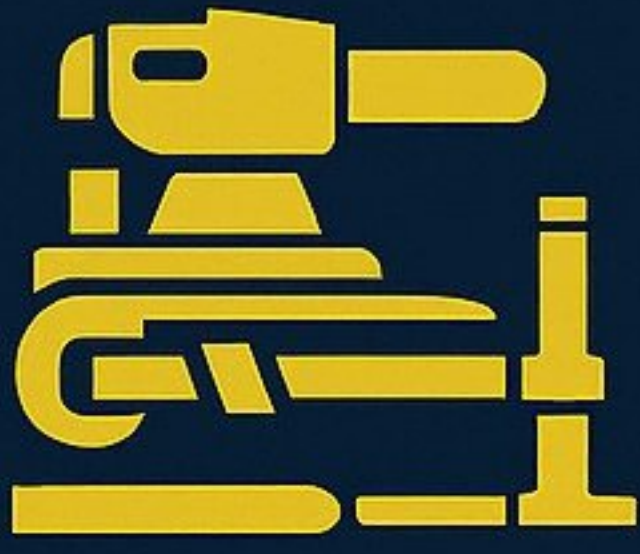
A structured process to report, record, and investigate incidents, near misses, and unsafe acts to prevent recurrence.

HAZARDS

- ⚠ Delayed or no reporting of incidents
- ⚠ Underreporting due to fear of blame or punishment
- ⚠ Recurrence of similar incidents due to lack of investigation
- ⚠ Loss of critical evidence when reporting is late
- ⚠ Miscommunication leading to incomplete reports
- ⚠ Inadequate root cause analysis
- ⚠ Failure to identify systemic or hidden hazards
- ⚠ Lack of corrective and preventive actions (CAPA)
- ⚠ Poor documentation and record-keeping
- ⚠ Negative safety culture discouraging openness

PRECAUTIONS

- ✓ Establish a clear incident reporting system (forms, hotline, app)
- ✓ Encourage reporting of all incidents and near misses
- ✓ Train workers on how and when to report incidents
- ✓ Promote a no-blame culture focused on learning
- ✓ Secure and preserve evidence immediately after incidents
- ✓ Conduct thorough root cause analysis (RCA)
- ✓ Implement corrective and preventive actions promptly
- ✓ Share lessons learned across all teams and sites
- ✓ Regularly review and update reporting procedures
- ✓ Establish and communicate an incident Management Policy



HORIZONTAL DRILLING / AUGER BORING SAFETY

Safe horizontal drilling or auger boring
under paved roads

HAZARDS

-  Striking or puncturing live buried pipelines or cables during drilling
-  Collapse of road surface above boring area
-  Equipment entanglement or mechanical failure inside borehole
-  Water, gas, or crude oil leakage from damaged utilities
-  Presence of undetected old or abandoned pipelines
-  Electrical shock from contact with underground power cables
-  Unstable soil causing borehole collapse
-  Lack of updated drawings or incorrect line coordinates
-  Poor supervision or lack of communication between drilling and line detection

PRECAUTIONS

-  Obtain valid Line Detection and UXO Clearance Certificates
-  Conduct survey and mark all underground utilities clearly on site
-  Use accurate detection tools (e.g., RD8200, GPR) during drilling
-  Ensure PTW and HITRA are reviewed and approved before work
-  Assign Line Detector / Supervisor for continuous
-  Maintain safe distance from existing buried pipelines
-  Stop drilling immediately if unexpected resistance or vibration occurs
-  Prepare and follow Emergency / Leak Response Plan
-  Ensure operator is trained and authorized for horizontal drilling